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2D Wave Diffraction Simulator

This project is a Python and Streamlit web app that simulates a two-dimensional wave traveling toward a wall with one or two narrow slits cut into it. The wave passes through the slits, spreads out, and forms a diffraction pattern on a screen line near the far edge of the grid. The user can switch between single-slit and double-slit mode and adjust the wavelength λ , the slit width w , the slit separation d , and the screen distance L . The simulation does not use the textbook interference formula to draw the picture. Instead, it integrates the 2D scalar wave equation

$$\frac{\partial^2 u}{\partial t^2} = c^2 \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right)$$

on a 640×320 grid using an explicit finite-difference time-domain (FDTD) leap-frog update.

The formula derived from the PDE,

$$u^{n+1}_{i,j} = 2u^n_{i,j} - u^{n-1}_{i,j} + r^2(u^n_{i+1,j} + u^n_{i-1,j} + u^n_{i,j+1} + u^n_{i,j-1} - 4u^n_{i,j}),$$

uses the values during the previous two time intervals and values of adjacent cells to update the current cell. The barrier is implemented in code as a hard wall ($u = 0$) carved with one or two openings, and a damping layer ensures that outgoing waves are absorbed and don't reflect back into the pattern.

The educational point of the simulation is to teach that the double-slit and single-slit formulae taught in class are not separate rules; rather, they are consequences of a single PDE.

The program contains no interference formulae in its update loop: The interference emerges on

their own. Sliding the wavelength up visibly spreads the fringes apart; sliding the slit separation up packs them closer together; switching to single-slit mode causes the central diffraction maximum to dominate, and widening that slit visibly narrows the central peak. A student can therefore observe relationships seen in class as direct consequences of the wave equation rather than as formulae learned through rote memorization.

The simulator can break down under these conditions:

1. When the slit width drops below roughly one wavelength, the wave can barely fit through and less amplitude gets transmitted
2. When the screen distance L is small, the geometry is in the Fresnel (near-field) zone, and measured spacings drift away from the prediction
3. When the diffraction angle $\theta = \sin^{-1} \frac{\lambda}{d}$ is too large, the small-angle approximation fails and the formula $\Delta y = \frac{\lambda L}{d}$ disagrees with the simulated fringe positions on screen.
4. The math behind FDTD only stays stable if $r \leq \frac{1}{\sqrt{2}}$ ($r = \frac{c\Delta t}{\Delta x}$). r is pinned to 0.45 so the slider UI cannot trigger this.
5. The screen is just a row of cells, so peaks always snap to whole-number positions.

That puts a hard floor on how small the percent error can get.

To test the simulation's accuracy, the app validates itself live, every run. In double-slit mode it detects peaks in the measured intensity, computes the mean fringe spacing, and compares it to $\frac{\lambda L}{d}$, and calculating a percent error. In single-slit mode it measures the full width at half maximum (FWHM) of the central maximum. I confirmed accuracy across a range of conditions; the representative results are below (All units are in grid units).

Mode	λ	w	d	L	% error
Double	8	10	40	460	0.5%
Double	10	10	40	460	10.9%
Double	8	15	30	350	17.1%
Double	10	15	30	350	11.0%
Double	8	8	35	460	10.4%
Single	8	18	—	460	15.5%
Single	8	15	—	400	2.1%
Single	8	9	—	350	1.0%

The percent error calculated in various settings

Works Cited

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